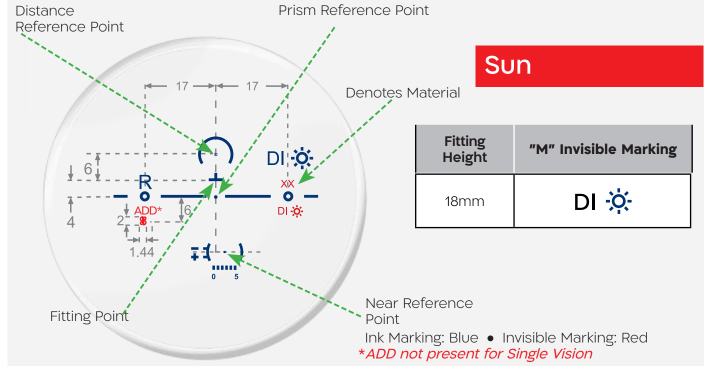
Driver Intelligence™

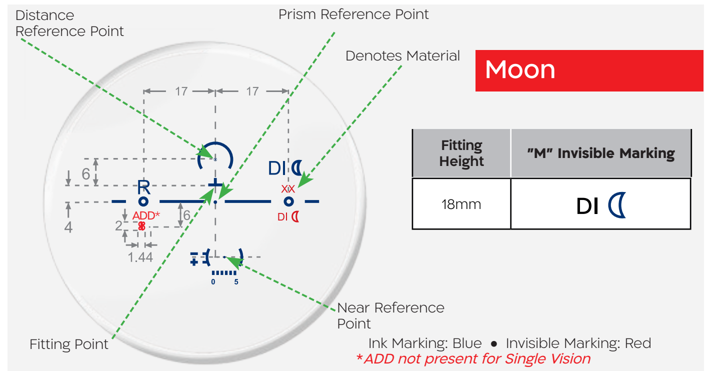
Availability

	Lens Product	Material	Premium AR Coating	Power Range	Add Power	Cylinder
Driver Intelligence™ - Progressive	Driver Intelligence™ Sun	Poly-polarized Gray/Brown	Glacier Sun™	-14.75 to +6.00	0.75 to 3.50	to -7.00
	Driver Intelligence™ - Moon	Poly clear	Glacier Expression	-14.75 to +6.00	0.75 to 3.50	to -7.00
Driver Intelligence - Single Vision	Driver Intelligence™ Sun	Poly-polarized Gray/Brown	Glacier Sun™	-14.75 to +6.00	N/A	to -7.00
	Driver Intelligence™ - Moon	Poly clear	Glacier Expression	-14.75 to +6.00	N/A	to -7.00

Coming soon: 1.67 index; as well as Color Enhancement lens materials.

TECHNICAL INFORMATION • PAL and Single Vision





Design Base Curves

	Frame Type	Flat Frame (0-13)			Flat Or Wrap Frame (0-30)				Wrap Frame (13-30)		
Driver Intelligence™ Progressive Sun & Moon	Base Curve	Base 0.50	Base 2.00	Base 3.00	Base 4.00	Base 5.00	Base 6.00	Base 7.00	Base 8.00	Base 9.00	Base 10.00
	Sphere Range	-14.75 to 0.25	-10.00 to 1.25	-8.50 to 2.25	-6.50 to 3.00	-6.50 to 4.00	-6.00 to 4.50	-5.50 to 5.00	-5.00 to 6.00	-3.00 to 6.00	-3.00 to 6.00
Driver Intelligence™ Single Vision Sun & Moon	Base Curve	Base 0.50	Base 2.00	Base 3.00	Base 4.00	Base 5.00	Base 6.00	Base 7.00	Base 8.00	Base 9.00	Base 10.00
	Sphere Range	-14.00 to -0.25	-12.00 to 1.25	-9.00 to 2.75	-6.50 to 3.00	-6.50 to 4.00	-6.00 to 5.00	-6.00 to 6.00	-6.00 to 6.00	-4.00 to 6.00	-3.00 to 6.00

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SHA-LBFLY-1117-DI-082823

shamir

Driver Intelligence™

The Ultimate Driving Vision Technology

At Shamir, we've always taken pride in making the extra effort to effectively "see through our users' eyes".

The Perfect Solution For Driving In All Conditions

Introducing a comprehensive solution that addresses varied visual needs and lighting conditions when driving, comprising of two pairs of lenses that deliver a 24 hour solution, for driving enthusiasts, professionals and everyday drivers.

The Designs

Shamir **Driver Intelligence**™ incorporates our most advanced technologies with unique benefits for PAL and Single Vision lenses.

Even better, both designs come with specifically paired coatings: **Driver Intelligence**™ **Sun** has our **Glacier Sun**™ AR coating and **Driver Intelligence**™ **Moon** has our world class **Glacier Expression**™ AR coating. We've already got your patients covered.

Shamir Driver Intelligence™ – A Multi-facted Solution

Shamir **Driver Intelligence**™ General Overview – Features vs. Visual Driving Challenges

Visual challenges experienced by drivers	Driver Intelligence™ features designed to address each challenge	Benefits of Driver Intelligence™ feature compared to a general purpose lens
Night myopia, higher prevalence among younger drivers	**Unique product design incorporates a myopic shift based on visual age**	Night myopia reduction as a function of age
Constant need to direct gaze to objects requiring over a 30° eye rotation (need for constant head rotation)	**Unique design incorporates wider and clearer visual fields in areas of interest**	Minimal head movement, more eye movement, improved ability to comfortably scan surroundings
Sun glare	**Coatings (Day) Shamir Glacier Sun™**	Sun glare reduction
Decline of visual acuity at night	**Coatings (Night) Shamir Glacier Expression™**	Improved acuity at night
Decline of contrast sensitivity at night		Improve contrast sensitivity at night
Increased reaction time at night		Faster reaction at night
Perceived hues create/increase glare	**Color enhancement*** *coming soon	Improved hue differentiation* during daylight *At NC effective filter zone
Perceived colors create/increase glare		Better color perception during daylight
High cognitive load while driving	**Expected overall benefits**	Less cognitive load experienced while driving

How To Order: Provide the following information*:

- Select the lens type: Shamir **Driver Intelligence**™ PAL or Shamir **Driver Intelligence**™ SV**
- Patient's prescription (prescription should be measured with a pantoscopic tilt of zero in the trial frame)
- Frame Data: A, B, ED & DBL or frame tracer file
- Fitting height & Mono PD (Distance & Near)
- Panoramic angle
- Pantoscopic tilt (patient should hold their head in a natural position and look straight during measurement)
- Vertex distance
- Preferred Base Curve: Obtained by clocking the base curve of the demo lens at the fitting point; = Enter into NOTES/SPECIAL INSTRUCTIONS.
- Frame type

Plano can be ordered in Shamir **Driver Intelligence™ SV Sun/ Moon but will return compensated values.*

***Sun and Moon must be ordered together as a 2 pair package.*

Night Myopia, More Prevalence In Younger Drivers

- The main cause of night myopia was found to be the accommodation shift occurring at low light levels[1].
- Night myopia > 0.75 D has been reported to affect 17% of randomly selected individuals aged 16-80 and 38% of those aged 16-25. Driving at night could create visual difficulties for some younger patients that a night myopic correction would eliminate[2].

1. Artal P, et al. Night Myopia Studied with an Adaptive Optics Visual Analyzer. 2012. PLoS ONE 7(7): e40239.

2. Fejer TP, Girgis R. Night myopia: implications for the young driver. 1992. Can J Ophthalmol 27: 172-176.

Driver Intelligence™

Lens Design Based on Elements of AI

Following the use of Big Data collection tools, and the Shamir Artificial Intelligence-tooled Engine for analyzing this data, an understanding was gained of the most important areas of the lens for optimal driving vision, enabling Shamir's R&D team to develop the most suitable lens design carefully adapted to the special visual needs of the driver. Its unique design ensures minimal head movement and greater freedom of eye movement. This in turn contributes to greater comfort and safety while driving, improved reaction times, and an improved ability to scan the surroundings, allowing for overall better driving control.

Among its other features, the lens is characterized (see description below in Fig. 1) as a soft, low aberration design, with excellent distance vision emphasizing sharp vision for the car mirrors. Clear vision for intermediate and close distances (22-36in) is enabled through the lens power profile that matches modern car interior settings. This high visual quality is achieved thanks to Shamir's leading lens design technologies that provide visual acuity, ergonomics, and comfort of vision.

Lastly, to provide greater visual acuity, the Shamir lens design takes into account calculations of myopic shift based on visual age. For far vision, the minus power at the upper part of the lens compensates for the phenomena commonly called night myopia. Night myopia, as described in the literature review section, affects about 20% of the population and is more common among young adults. Thanks to its **Continuous Design Technology**, Shamir can provide gradual prescription correction per the patient's **Visual Age**, without compromising daylight vision quality (integrating myopic shift only in the lenses designed for use under low light conditions).

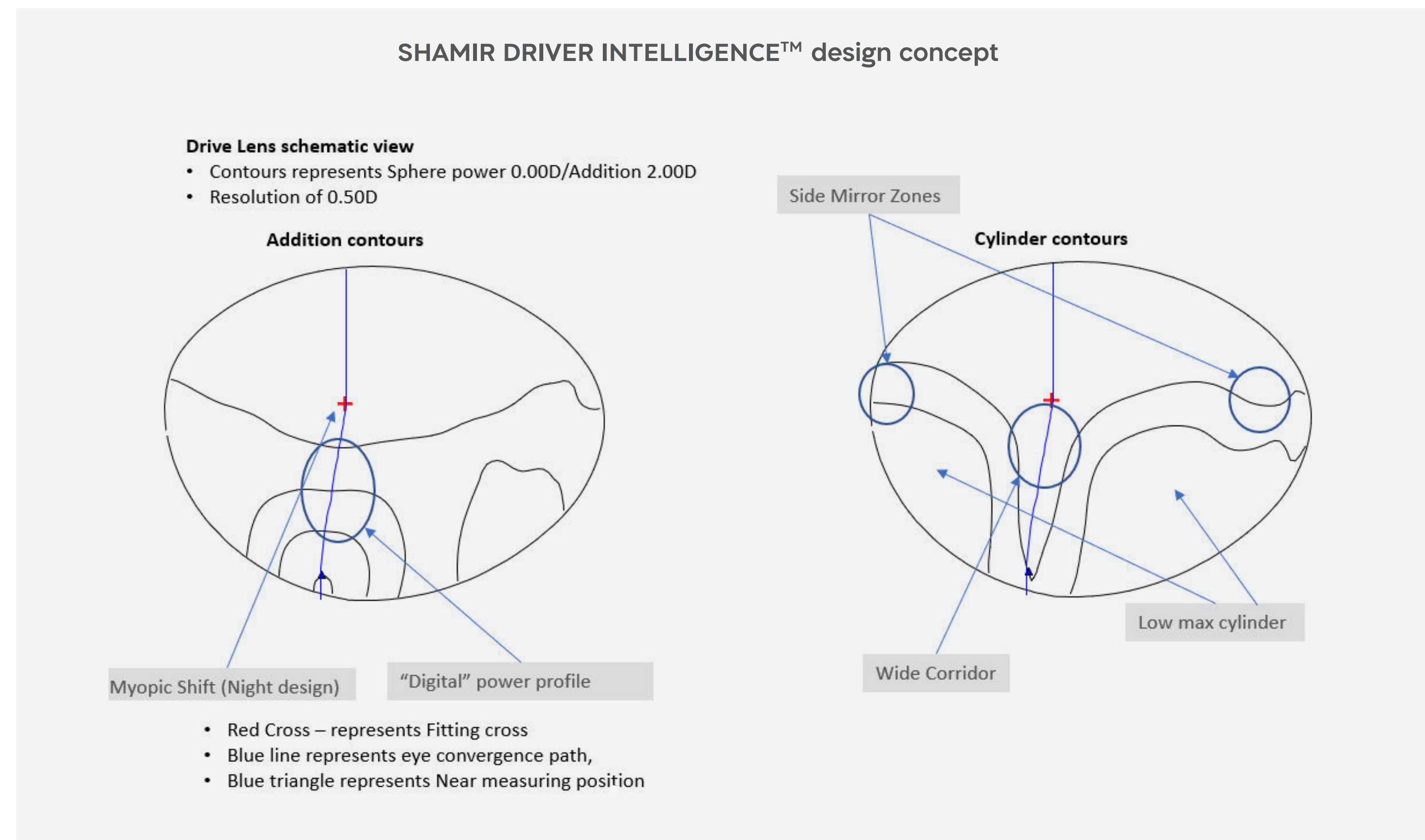

Figure 1. *SHAMIR DRIVER INTELLIGENCE design concept fitted to the driver visual needs and modern vehicle interior settings: the design has low maximum level of aberrations (cylinder), wide corridor, wide far field of view and a power profile better matched to driving properties.*